

Block vs Inline Elements, and Div & Span

Let's get
started!



Today's Topics

❖ What are Block and Inline

❖ Layout Behavior

Elements

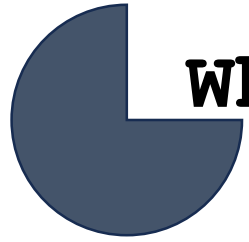
❖ `<div>` and `<span>` Usage

❖ Differences Between Them

❖ Semantic Importance

❖ Examples of Each Type

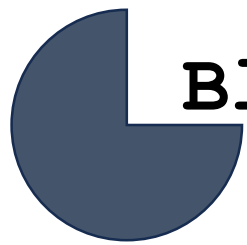




What are HTML Elements?

- HTML elements are building blocks of a webpage.
- Each element defines a **type of content** and **how it behaves** in layout.
- Elements are categorized into two main types:
 - **Block-level elements**
 - **Inline elements**
- Understanding these types is essential for **designing page structure**





Block-Level Elements

- Start on a new line and take up the full width available.
- Stack vertically.
- Useful for creating structure or layout sections.

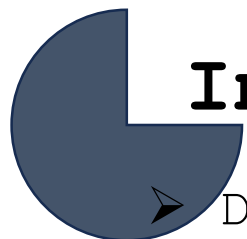
Examples:

```
<div>, <p>, <h1>-<h6>, <ul>, <ol>, <li>, <section>, <article>,
```

Behavior:

Automatically breaks to a new line before and after the element.





Inline Elements

- Do not start on a new line.
- Take up only as much width as needed.
- Appear within block elements.

Examples:

```
<span>, <a>, <img>, <strong>, <em>, <label>,
```

```
<input>
```

Behavior:

- Inline with surrounding content.
- Cannot contain block-level elements inside.





Block vs Inline – Key Differences

Feature	Block Element	Inline Element
Layout	Takes full width	Takes only content width
Starts on new line	Yes	No
Nested inside	Can contain any element	Only inline or text
Examples	<code><div>, <p></code>	<code>, <a></code>

➤ Choosing the correct type affects page flow and design flexibility.





Visual Comparison

Block Example:

```
<p>This is a paragraph.</p>  
<h2>This is a heading</h2>
```



This is a paragraph.

This is a heading

Inline Example:

```
<span>This is inline</span>  
<a href="#">Click here</a>
```



This is inline [Click here](#)

➤ Block elements stack vertically; inline elements line up horizontally.





The `<div>` Tag

- Short for "division".
- A generic block-level container.
- Commonly used for grouping content and applying CSS styling.

```
<div class="container">  
  <p>Paragraph inside a div</p>  
</div>
```

- `<div>` has no semantic meaning; it's purely structural.





The `<span>` Tag

- An inline-level container.
- Used to style or highlight a portion of text within a line.
- Does not affect layout but is useful for applying CSS/JS.

```
<p>This is <span style="color:red;">important</span> text.</p>
```

- Like `<div>`, `<span>` is non-semantic but ideal for small inline styli





When to Use `<div>` and `<span>`

Use Case	Use <code><div></code>	Use <code></code>
Grouping block content	Sections, cards, containers	Words/phrases within a paragraph
Layout divisions	Columns, grids	Styling or scripting inline text
Styling	Entire sections	Specific text chunks

- Use semantic tags when possible, and reserve `<div>` and `<span>` for custom styling or JS hooks.





Nesting Block and Inline Elements

✓ Allowed:

```
<div><p><span>Text</span></p></div>
```

✗ Invalid:

```
<span><div>Not allowed</div></span>
```

- Inline elements should not contain block-level elements inside.





CSS Styling with Div & Span

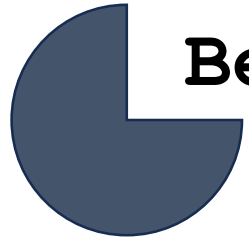
- `<div>` often used with IDs or classes to apply layout styling.
- `<span>` allows inline color, font, size changes.

Example:

```
<div class="card">...</div>  
<span style="font-weight:bold;">Bold text</span>
```

- Ideal for responsive design, grid systems, and component-based styling.





Best Practices

- Use semantic elements like `<section>`, `<article>` when structure has
- Use `<div>` for layout containers; avoid excessive nesting.
- Use `<span>` sparingly for targeted text styling.
- Avoid using `<div>` or `<span>` without purpose or content.
- Cleaner markup = Better SEO, Accessibility, and Maintenance.





